

✓ EC2 + Git + Docker + Nginx – FULL COMMANDS CHEATSHEET

☐ 1. Update System & Install Git

```
sudo yum update -y
sudo yum install git -y
```

☐ 2. Configure Git (First Time Only)

```
git config --global user.name "your-github-username"
git config --global user.email "your-email@example.com"
```

☐ 3. Clone Your Repository

```
git clone https://github.com/<username>/<repo>.git
```

Move inside the project folder:

```
cd <repo>
```

☐ 4. If Git Folder Was Missing (Re-initialize Git)

```
git init
git remote add origin https://github.com/<username>/<repo>.git
```

Check remote:

```
git remote -v
```

☐ 5. Add All Files to Git

```
git add .
```

☐ 6. Commit the Files

```
git commit -m "Initial commit"
```

☐ 7. Push to GitHub (First Time: with -u for upstream)

```
git push -u origin main
```

☐ 8. If Push Fails Due to Token — Set Personal Access Token

```
git remote set-url origin https://<username>:<TOKEN>@github.com/<username>/<repo>.git
```

Try pushing again:

```
git push -u origin main
```

☐ 9. Create your *Dockerfile* inside EC2

Example:

```
vim Dockerfile
# 1. Use official nginx image as base
FROM nginx:latest

# 2. Remove the default nginx webpage
RUN rm -rf /usr/share/nginx/html/*

# 3. Copy your project files into nginx web directory
COPY . /usr/share/nginx/html/

# 4. Expose port
EXPOSE 80

# 5. Start Nginx (default command)
CMD ["nginx", "-g", "daemon off;"]
```

(Press **i** to edit, **ESC** → **:wq** to save)

☐ 10. Build Docker Image

```
sudo docker build -t myapp .
```

List images:

```
sudo docker images
```

☐ 11. Run Container (Expose Port 80)

```
sudo docker run -d -p 80:80 myapp
```

Check running containers:

```
sudo docker ps
```

☐ 12. Stop/Remove Containers

Stop:

```
sudo docker stop <container_id>
```

Remove:

```
sudo docker rm <container_id>
```

☐ 13. Remove Docker Image

```
sudo docker rmi myapp
```

☐ 14. Install Nginx (Optional)

```
sudo yum install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

Check status:

```
systemctl status nginx
```

☐ 15. If using Nginx Reverse Proxy

Open Nginx config:

```
sudo vim /etc/nginx/nginx.conf
```

Test config:

```
sudo nginx -t
```

Restart:

```
sudo systemctl restart nginx
```

☐ 16. Verify Application

```
curl localhost
```

Or open browser:

```
http://<EC2-public-IP>
```

☐ 17. Basic Linux Commands

```
pwd      # Show current directory
ls -la   # List all files (incl hidden)
cd folder # Move inside folder
cd ..    # Go back
rm file  # Delete file
rm -rf dir # Delete folder and content
```

❑ 18. Fixing COMMON ERRORS

◆ "current commit information not captured"

Run:

```
git rev-parse --is-inside-work-tree
```

If output is fatal → initialize git:

```
git init
```

◆ "403 Permission Denied"

Reset remote with token:

```
git remote set-url origin https://<user>:<TOKEN>@github.com/<user>/<repo>.git
```

◆ "Nothing to commit" but new files exist

```
git add .  
git commit -m "message"  
git push
```